

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Calculate the percentage composition of carbon dioxide(CO_2) (Atomic weight of C=12 and O is 16.) (CO1)

Q.24 Explain proximate and ultimate analysis of coal. (CO2)

Q.25 Write factors affecting adsorption of gases on solids. (CO4)

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Roll No.

2nd Sem / Ceramic

Subject : Chemistry Applications

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Symbol N stands for_____ (CO1)

- a) Calcium b) Nitrogen
- c) Carbon d) Copper

Q.2 Composition of producer gas is_____ (CO2)

- a) $\text{CO} + \text{CH}_4$ b) $\text{CO} + \text{N}_2$
- c) $\text{CO} + \text{H}_2$ d) $\text{CH}_4 + \text{H}_2$

Q.3 The point at which degree of freedom is zero is called _____ (CO3)

- a) Vapour point b) Eutectic point
- c) Sublimation point d) Triple point

Q.4 What is Atomic mass for Carbon? (CO1)

- a) 12 b) 13
c) 14 d) 16

Q.5 In Gibbs phase rule degree of freedom is represented by _____ (CO3)

- a) C b) P
c) F d) none of these

Q.6 The property of colloids include

- a) tyndall effect b) vaporization
c) condensation d) none of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define atomic mass. (CO1)

Q.8 Calorific value of fuel is determined by which instrument? (CO2)

Q.9 Petrol is a solid fuel(T/F) (CO2)

Q.10 Define the phenomena of adsorption. (CO4)

Q.11 Define Ceramics. (CO6)

Q.12 Define Tyndall Effect. (CO5)

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SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write down the formula of following compounds (CO1)

- a) Calcium Carbonate
b) Hydrogen Floride

Q.14 Write essentials of a chemical equation. (CO1)

Q.15 Define combustible and non-combustible constituents of coal. (CO2)

Q.16 Write any four uses of bio-gas. (CO2)

Q.17 Explain the concept of fusion and vaporization curve. (CO3)

Q.18 Define uniary phase, binary phase, ternary phase with examples of each. (CO3)

Q.19 Write any four differences between physiosorption and chemisorption. (CO4)

Q.20 Explain the terms deflocculation and coagulation. (CO5)

Q.21 Define paint. Write down any 4 advantages of this organic coating. (CO6)

Q.22 Write down chemical composition and application of borosilicate (CO6)

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